CASA DE LÂNGĂ LAC

Visual: project identity graphic generated from the supplied project list. Official project images are available through the source links below.

Why is this project a best practice?

Challenge

The project had to provide high thermal comfort through cold winters and warm summers while sharply reducing operational energy. The main design challenge was to coordinate a high-performance envelope, efficient services, renewable energy and healthy materials without compromising the architectural concept, construction quality or everyday usability.

Solution

The response combines prefabricated timber-frame construction with 30 cm wall insulation and 40 cm roof insulation; mechanical ventilation with heat and moisture recovery, plus exceptional airtightness verified by testing; and ground-source heat pump, underfloor heating, ceiling cooling, triple glazing and a photovoltaic system sized to cover annual demand. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The core package - demand reduction first, airtight construction, efficient low-temperature systems, ventilation and on-site renewables - is transferable to other Romanian homes. Replication requires climate-specific energy modelling, careful thermal-bridge and airtightness detailing, qualified installation, commissioning and clear operating guidance for residents.

Key facts and figures

Location	Pantelimon, Ilfov County, Romania
Building use / function	Single-family dwelling

Project type (new build, refurbishment, extension)	New build; certified Passive House Premium
Plot area / Floor area	236 m² gross; approximately 170 m² usable
Year / phase	Constructed 2020-2022; occupied and certified
Climate zone	Temperate continental; cold winters and warm summers
Thematic focus / innovation	Timber-frame Passive House Premium; airtight envelope; heat recovery; ground-source heat pump; photovoltaic self-generation
Investor / building owner	Private owners: Ion and Daniela Iosif
Architect	Not publicly identified in the consulted sources
Other involved stakeholders	Litarh (prefabricated timber structure); Eco Instal / Ovidiu Țifui (building services); Adriana Sîngeap (energy and certification); Kulttur/Internorm (high-performance windows)
Contact person	Litarh / Romanian Passive House Association; no individual public project contact provided
Further information	https://revistadinlemn.ro/2025/10/30/casa-de-langa-lac-pasiva-premium-isi-deschide-portile-in-cadrul-international-passive-house-open-days-2025/ https://video.revistadinlemn.ro/playlist/casa-de-langa-lac-drumul-spre-casa/

Casa de lângă lac is a new build; certified passive house premium project in Pantelimon, Ilfov County, Romania. It functions as single-family dwelling. Its main best-practice contribution is Romania's first publicly documented timber-frame dwelling certified at Passive House Premium level. The documented strategy brings together prefabricated timber-frame construction with 30 cm wall insulation and 40 cm roof insulation, mechanical ventilation with heat and moisture recovery, plus exceptional airtightness verified by testing, and ground-source heat pump, underfloor heating, ceiling cooling, triple glazing and a photovoltaic system sized to cover annual demand. Reported occupied performance includes very low heating/cooling demand; the source cites no more than about 5 kWh per day for the occupied 170 m² area. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - Prefabricated timber-frame construction with 30 cm wall insulation and 40 cm roof insulation.
2. Integrated systems and operation - Mechanical ventilation with heat and moisture recovery, plus exceptional airtightness verified by testing.
3. Wider value - Ground-source heat pump, underfloor heating, ceiling cooling, triple glazing and a photovoltaic system sized to cover annual demand.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.